

Customer Churn Dataset — Auto Analysis

Business Analytics Report

Project: Customer Churn Dataset

Dataset: customer_churn_dataset.csv

Analysis date: 2026-09-25

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1. Executive Summary

This report analyzes **customer_churn_dataset.csv** (50 rows, 14 columns) to answer the business question: *A synthetic dataset of 2,025 subscription customers with 17 columns covering demographics, subscription plan, contract type, payment method, tenure, spending, purchase activity, support tickets, and satisfaction scores. The target column "churn" shows whether the customer left (1) or stayed (0). It includes some missing values, duplicate rows, outliers, and invalid dates for testing data cleaning. Suitable for churn classification and revenue regression..*

Key verified findings

- The original dataset contained 0 missing value(s) and 0 duplicate row(s), which have since been cleaned.
- A Logistic Regression model was trained to predict **Churn** (classification). The model correctly classifies about 100.0% of cases (Accuracy).

Model reliability, in plain terms: The model correctly classifies about 100.0% of cases (Accuracy). F1 score (balancing precision and recall) is 1.000. ROC-AUC is 1.000 (1.0 = perfect separation, 0.5 = no better than chance).

Top recommendations

- The most common observed outcome for Churn is 'No' (5 of 10 test cases).
- Among customers, the 'Monthly Charges: Lower' group shows the highest observed rate (58.8%, n=17), compared to 'Monthly Charges: Higher' (41.2%, n=17).

2. About the Dataset

`customer_churn_dataset.csv` contains 50 rows and 14 columns.

Business question: A synthetic dataset of 2,025 subscription customers with 17 columns covering demographics, subscription plan, contract type, payment method, tenure, spending, purchase activity, support tickets, and satisfaction scores. The target column "churn" shows whether the customer left (1) or stayed (0). It includes some missing values, duplicate rows, outliers, and invalid dates for testing data cleaning. Suitable for churn classification and revenue regression.

Note: the project description mentions Revenue, Target, but no corresponding column was found in this dataset — this report cannot speak to that directly.

Column dictionary

Column	Type	Missing	Notes
Customer Id	Identifier	0.0%	typical range 22.00–55.00
Age	Category	0.0%	most common: Male
Gender	Numeric	0.0%	typical range 2.00–60.00
Tenure	Numeric	0.0%	typical range 49.00–99.00
Monthly Charges	Numeric	0.0%	typical range 98.00–5,940.00
Total Charges	Category	0.0%	most common: Month-to-month
Contract Type	Category	0.0%	most common: Bank transfer
Payment Method	Category	0.0%	most common: DSL
Internet Service	Category	0.0%	most common: Yes
Online Security	Category	0.0%	most common: No
Tech Support	Category	0.0%	most common: Yes
Streaming Tv	Category	0.0%	most common: No
Streaming Movies	Category	0.0%	most common: No
Churn	Category	0.0%	most common: No

3. Data Quality & Cleaning

Original dataset quality (as uploaded)

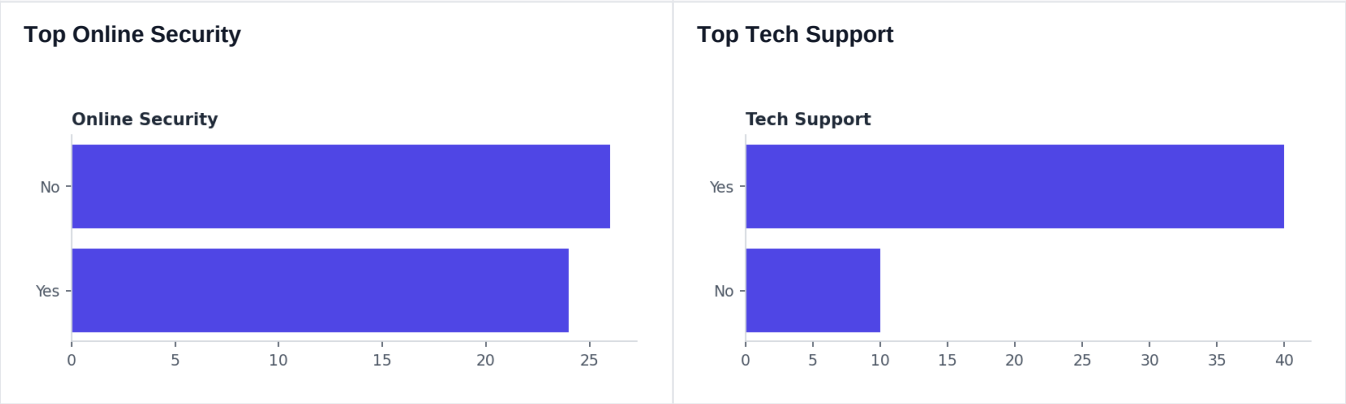
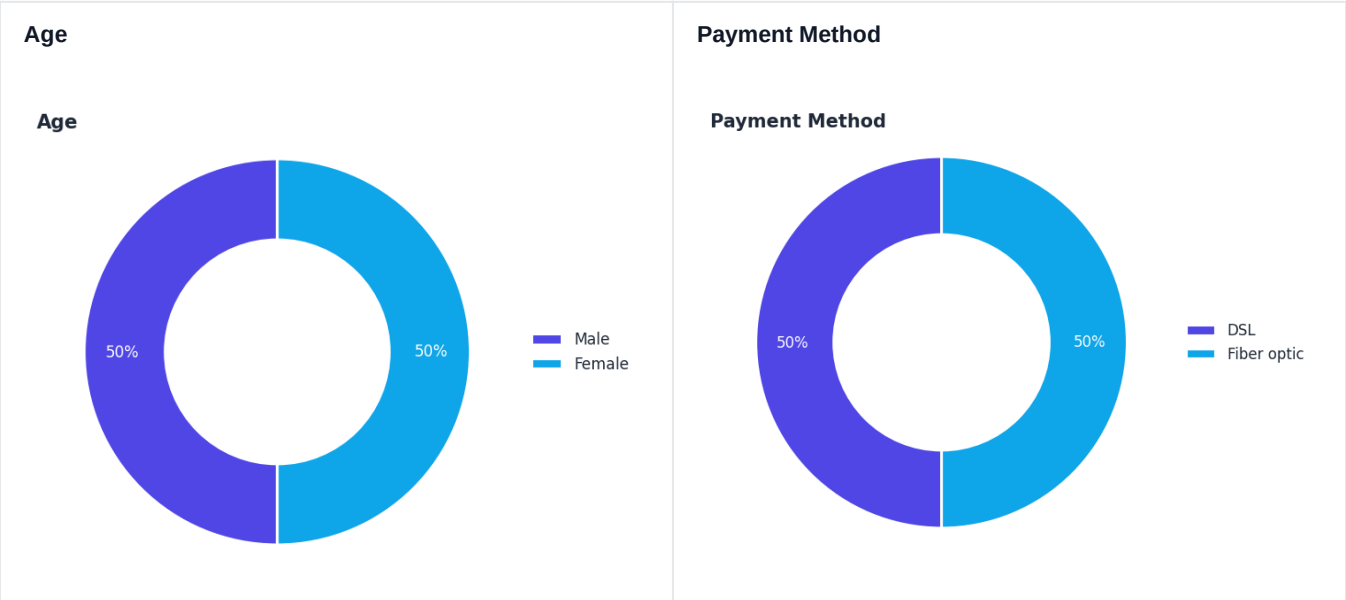
Total rows	Missing cells	Duplicate rows	Issues detected
50	0	0	0

Cleaning performed

The dataset was marked cleaned, but no specific actions were logged.

4. Key EDA Findings

Total Rows	Churn Rate	Total Gender	Total Tenure
50	48.0%	1,371	3,936



5. Modeling Approach

A **classification** model was trained to predict **Churn**, using 12 feature(s).

Features used

Age, Gender, Tenure, Monthly Charges, Total Charges, Contract Type, Payment Method, Internet Service, Online Security, Tech Support, Streaming Tv, Streaming Movies

Features excluded, and why

Column	Reason excluded
Customer Id	Identifier column (e.g. an ID) — carries no predictive signal.

Reviewed, not excluded: Internet Service (77.0% correlated) — no evidence these are derived from the target, so they were kept as legitimate predictors, but worth a manual sanity check.

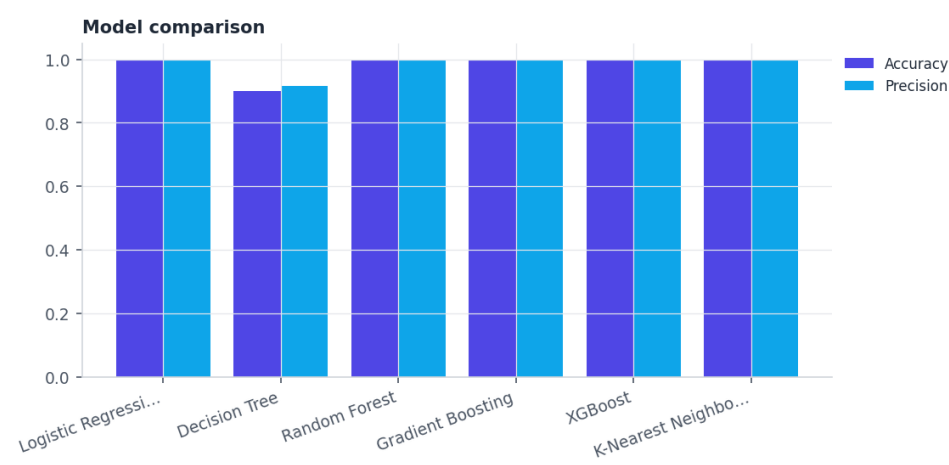
Technical details

Algorithm: Logistic Regression · Preprocessing: 3 numeric + 9 categorical feature(s), scaled/encoded · Train/test split: 20.0% held out · Cross-validation folds: 5 · Random seed: 42

6. Model Performance

Model comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	1.0000	1.0000	1.0000	1.0000	1.0000
Decision Tree	0.9000	0.9167	0.9000	0.8990	0.9000
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000
Gradient Boosting	1.0000	1.0000	1.0000	1.0000	1.0000
XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000
K-Nearest Neighbors	1.0000	1.0000	1.0000	1.0000	1.0000

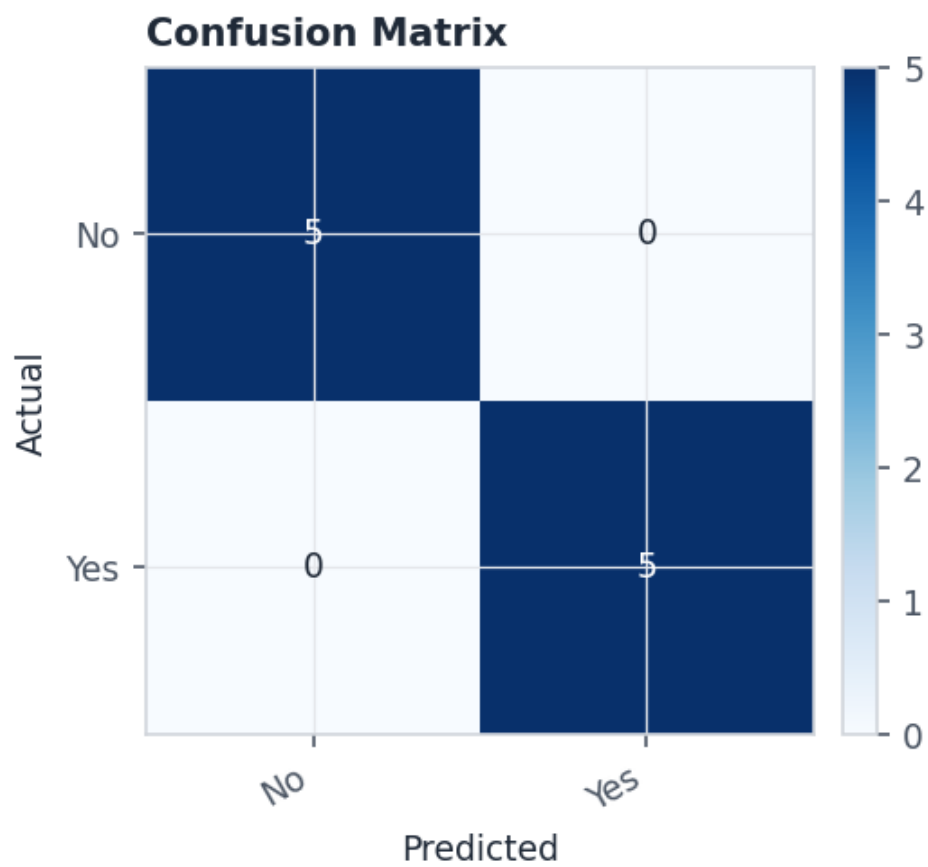


In plain language

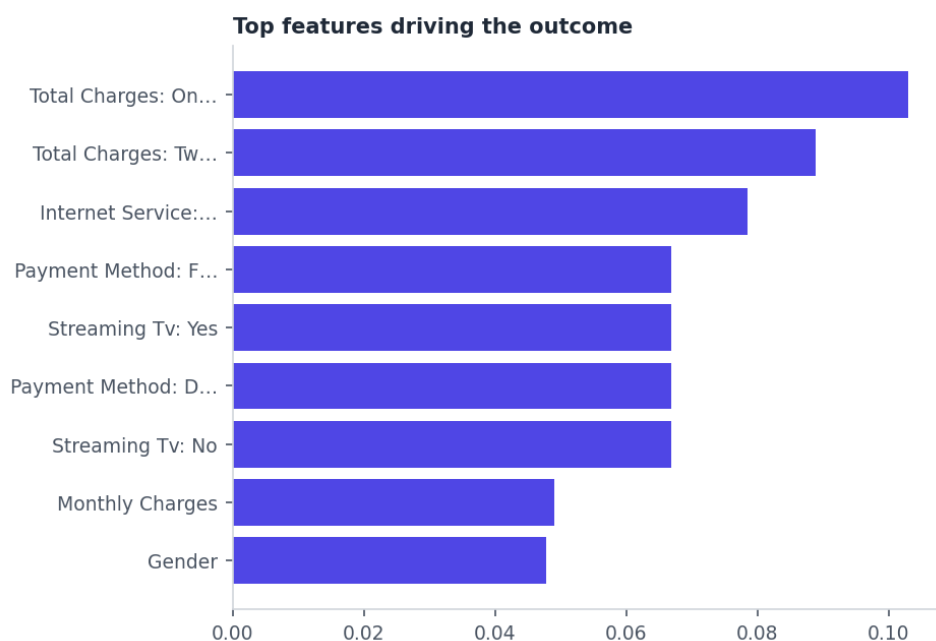
- The model correctly classifies about 100.0% of cases (Accuracy).
- F1 score (balancing precision and recall) is 1.000.
- ROC-AUC is 1.000 (1.0 = perfect separation, 0.5 = no better than chance).

Compared against a trivial baseline (F1 Score = 0.3333): this model clearly outperforms it.

Confusion Matrix



7. What Drives the Outcome



Feature	Relative importance
Total Charges: One year	10.3%
Total Charges: Two year	8.9%
Internet Service: No	7.8%
Internet Service: Yes	7.8%
Payment Method: Fiber optic	6.7%
Streaming Tv: Yes	6.7%
Payment Method: Dsl	6.7%
Streaming Tv: No	6.7%
Monthly Charges	4.9%
Gender	4.8%

8. Future Outlook

- The most common observed outcome for Churn is 'No' (5 of 10 test cases).
- Among customers, the 'Monthly Charges: Lower' group shows the highest observed rate (58.8%, n=17), compared to 'Monthly Charges: Higher' (41.2%, n=17).

Group	Observed rate	Group size
Monthly Charges: Lower	58.8%	17
Monthly Charges: Middle	43.8%	16
Monthly Charges: Higher	41.2%	17

9. AI Business Insights AI-generated

AI insights are unavailable right now (gemini: Gemini rate limit exceeded — too many requests. Try again shortly.; ollama fallback: The AI service is currently unavailable.). The rest of this report contains verified analytical results only.

10. Limitations

- This dataset has only 50 rows — results and model performance should be treated as preliminary; a larger sample would give more reliable estimates.
- Column(s) with high correlation to the target were retained but should be manually verified as legitimate, known-in-advance predictors: `internet_service`.
- Cross-validation scores varied by ± 0.049 across folds — treat the headline metric as an estimate with this much natural uncertainty, not an exact figure.
- Model predictions are estimates based on historical patterns and may not hold if underlying business conditions change. They should inform, not replace, domain expertise.

11. Appendix

Full model metrics

Metric	Value
Accuracy	1.0
Precision	1.0
Recall	1.0
F1 Score	1.0
ROC-AUC	1.0
Cv Mean	0.9754
Is Weak	False

Preprocessing

Setting	Value
Test Size	0.2
Cv Folds	5
Random seed	42
Model version	1

Hyperparameters

Parameter	Value
Random Seed	42

Dataset statistics

Statistic	Value
Rows	50
Columns	14